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Multimodal Medical Image Fusion using Hybrid Layer Decomposition with CNN-based Feature Mapping and Structural Clustering

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***Abstract*—Multimodal medical image fusion aims to integrate multisensory medical data into a single image for better prediction of the diagnostic details. This paper presents a feature level multimodal medical image fusion with the use of two-scale $\ell_1 - \ell_0$ hybrid layer decomposition scheme to maximize the structural details with suppression of significant noise and artifacts. The proposed method utilizes a convolutional neural network (CNN) with consistency verification and structural patch clustering (fuzzy c-means based) for the decomposed base and detail layer fusion, respectively. At first, a color space transform is used to separate the luminance and chrominance components extracted from both the source images and the luminance part of each image is decomposed. In the second step, a pre-trained CNN model is used to extract the prominent features from each decomposed base layer component. For the output feature map, a regional energy-based activity measure is computed and generated a fusion score, which is refined in the consistency verification step to optimize the weight map for fusing the base layers. The two-scale detail layers are merged by clustering based pre-learned dictionary for efficiently mapping of structural details of the layers. The color components associated with both the images are also combined using a color saliency measure. Finally, the fused base layer, detail layers, and color components are merged to get the resultant fused image. The experimental results justify the superiority of the proposed approach in terms of both subjective and objective assessments. The performance of the proposed scheme is tested on a large multimodal medical dataset of MR-SPECT, MR-PET, and CT-MR neurological images. Furthermore, the proposed approach shows better results compared to the other state-of-the-art approaches with improved fusion quality and computational performances.**

***Index Terms*—Medical image fusion, Clustering, Principle component analysis, Guided filter, $\ell_1 - \ell_0$ layer decomposition.**

I. Introduction

In recent years, many medical imaging techniques are being extensively used for visualizing the complementary information. The aim of medical image fusion is to combine the information captured from these multimodal images in a single composite image and improve the diagnostic accuracy. Normally, no single imaging modality can reflect a complete tissue information hence, the problem of information overloading occurs. The motivation is to preserve the maximum diagnostic details from the input images with perceptible visual quality without introducing any spatial and phase distortion.

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Over the last few years, many research works have been reported on multimodal medical image fusion [1]. The fusion techniques are broadly categorized in pixel [2], feature, and decision level. Usually, these levels of abstraction for image fusion are designed in spatial and transform domain. However, spatial domain methods suffer from poor contrast and spatial localization. In this context, different multiscale decompositions as discrete wavelet transform (WT) [3], curvelet transform (CVT) [4], stationary wavelet transform (SWT) [5], dual-tree complex wavelet transform (DTCWT) [6], contourlet transform (CT) [7], nonsubsampled contourlet transform (NSCT) [8], [9] and nonsubsampled shearlet transform (NSST) [10] etc have been introduced to design an effective platform which provide better localization of an image contour and texture details with superior visual quality. The NSCT based approaches show quite acceptable fusion results under different combinations of fusion rules. Yang et al. have proposed a fuzzy adaptive pulse-coupled neural network (PCNN) in NSCT domain [11]. To overcome the limitations of NSCT, NSST based fusion rules are framed and look promising for different image processing applications but lack in pixel contrast and preservation of tiny edges [12], [13].

An edge-preserving decomposition technique as hybrid $\ell_1 - \ell_0$ prior based layer decomposition [14] considers two separate regularization parameters for base and detail layer to retain the week structures and sharp edges. This model is able to avoid the halo artifacts and provide better visual quality. A unified fusion framework is proposed to fuse the multimodal images using wavelet transform and generalized homomorphic filtering [15] in which the control parameters are estimated using grey-wolf optimization. This provides significant results but suffers from high computational cost. The guided filtering based image fusion [16] has also become popular as it retains the decent spatial consistency, but suffers from spatial smoothing or staircase effects at image boundary. Currently, deep learning has been efficiently used in several applications of image processing such as image segmentation, classification, feature mapping, and decision making. A multimodal medical image fusion scheme based on the convolutional neural network (CNN) [17] has taken an important role in the processing steps for better performance of pixel to decision level fusion [18]. The major drawbacks of such methods are execution complexity and the requirement of extensive data for model training [19]. Furthermore, a convolutional sparsity-based morphological component analysis (CS-MCA) based fusion approach is proposed [20] which shows better results,

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but still, there is a scope to improve the performance for functional imaging modality.

In this paper, a novel fusion model with CNN and PCA clustering is proposed for multimodal medical neurological images. The proposed framework is able to efficiently capture the spatial information, maintain the spatial consistency and suppresses noise and artifacts. The major contributions of the proposed framework are featured below:

- A novel two-scale composite $\ell_1 - \ell_0$ layer decomposition model is employed to preserve the desired edges and intensity variations at each scale.
- To generate pixel activity and fusion weight map, a CNN based method followed by consistency verification (binary thresholding, iterative morphological filtering and weighted guided filtering) is employed due to the low accuracy of medical images.
- Fuzzy c-means clustering based local content mapping method is applied to capture the complex detail layer structures followed by PCA based subspace learning.
- The effectiveness of the proposed framework is demonstrated using extensive experimental results and comparisons made using some state-of-the-art fusion techniques for both the gray and color medical images.

The remaining part of the paper is structured as follows: Section II reviews the brief discussion on related works. Section III presents the proposed fusion model with salient steps for implementation. Section IV illustrates the simulated experiments with their results and finally, conclusions are summarized in Section V.

II. Preliminaries

A. Hybrid $\ell_1 - \ell_0$ Decomposition

The composite employment of ℓ_1 and ℓ_0 sparse regularization parameters, the presented image decomposition approach is structured as discussed in [14]. The ℓ_1 gradient prior term retains the image edges very well, but due to over-smoothing nature, it shows poor preservation of the structural information. However, the ℓ_0 gradient prior term holds clear piecewise constant property, hence greatly preserves the structural details. Firstly, apply the $\ell_1 - \ell_0$ priors on the input image to get the first scale decomposed components formulated as follow,

$$\left.\begin{array}{c} b_1 = model_{s=1}^{\ell_1,\ell_0}(I_X) \\ d_1 = I - b_1 \end{array}\right\} \tag{1}$$

where s indicates the scale while I, b_1, and d_1 show the input image, first scale base and detail layer, respectively. The equation for an $\ell_1 - \ell_0$ optimization model $model_{s=1}^{\ell_1,\ell_0}$ is given as:

$$min_b \sum_{i=1}^{N} (I^i - b^i)^2 + \lambda_1 \sum_{j=x,y} \left|\partial_j b^i\right| + \lambda_2 \sum_{j=x,y} F(\left|\partial_j d^i\right|) \tag{2}$$

where I^i is the ith pixel index of input image (I), b and d refer to the base layer and detail layer obtained by $(I - b)$. $\partial_j \in \mathcal{R}^{2N \times N}$ is the partial derivative operator for base and detail layers along $j = x$ or y directon, and N is the total pixel count. Variables λ_1 and λ_2 are the two regularization parameters with respect to the two gradient sparsity terms ℓ_1 and ℓ_0, respectively. The first term specifies the minimization of squared error between original image and decomposed base layer in order to correctly replicate the main information. The second term indicates ℓ_1 gradient sparsity prior (edge-preserving) imposed on base layer to suppress the outliers and third term indicates ℓ_0 gradient sparsity prior (structure-preserving) imposed on detail layer to suppress the artificial and weak textures with a function $F(.)$ which indicates,

$$F = \begin{cases} 1, & x \neq 0 \\ 0, & else \end{cases} \tag{3}$$

The regularization parameters λ_1 and λ_2 control the degree of smoothness for the base and detail layer, respectively. These regularization parameters are largely dependent on the pixel distribution present in the source image at first scale decomposition. Thus, the parameters should be low for detail layer while for the base layer it must be higher. The variation in statistical behaviour lies more at the first stage of decomposition and reduces as decomposition level increases. For the second scale decomposition, the model in Eq. (2) is now applied in layer b_1 to preserve the textural details transferred onto it during first scale decomposition and the remaining part will contain only local brightness. The $model_{s=2}^{\ell_1}$ can be expressed as:

$$min_b \sum_{i=1}^{N} (b_1^i - b^i)^2 + \lambda_3 \sum_{j=x,y} \left|\partial_j b^i\right| \tag{4}$$

$$\left.\begin{array}{c} b_2 = model_{s=2}^{\ell_1}(b_1) \\ d_2 = b_1 - b_2 \end{array}\right\} \tag{5}$$

where λ_3 is a controlling parameter for final base layer b_2. λ_3 and works as weight for ℓ_1 gradient sparsity prior to preserve stronger gradients of the base layer while providing piecewise smoothing effect. To guide the final base layer components weighted guided filtering is performed on b_2 with b_1 to restore the spatial consistency in sharp boundaries. The reconstruction is done by summing all three layers d_1, d_2 and b_2, represented as,

$$I_{Rec} = d_1 + d_2 + b_2 \tag{6}$$

The hybrid composition of these two priors (ℓ_1 and ℓ_0 based sparsity parameters) shows an effective and efficient decomposition model shown in Fig. 1. To achieve edge-aware filtering for base layer, a weighted guided filter (WGF) [21] is used which enhances the image features by removing the halo artifacts.

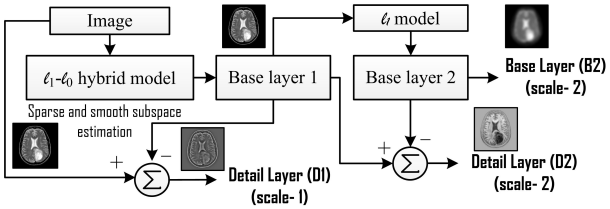


Fig. 1. A two-scale hybrid layer decomposition

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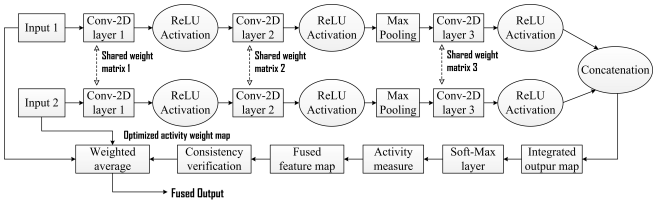


Fig. 2. Flow chart of CNN based fusion framework

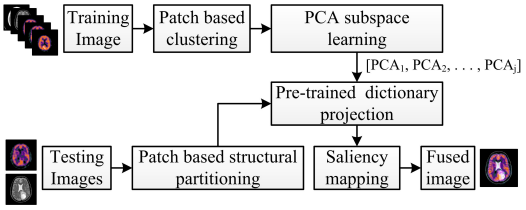


Fig. 3. Flow chart of the PCA clustering based fusion

B. Convolutional Neural Network

Nowadays, a CNN based method has become the majorly accepted model for multimodal medical input that is used to estimate the important visual features and corresponding score map. A CNN is a trainable feed-forward neural network having multiple stages corresponds to a certain number of feature maps. A Siamese based CNN architecture [18] used in the proposed work for feature mapping and base layer fusion has two channels which share similar weights and bias. This is also computationally easy to be trained compared to other networks. The pre-trained CNN model has three convolutional layers followed by non-linear activation function i.e. ReLu for each layer and one max-pooling layer. In the proposed fusion framework, a CNN model [18] is trained with 2 million patches of real and Gaussian filtered images taken from ImageNet dataset [22] using a deep learning platform Caffe [23].

The training images are partitioned into several patches of 16×16 size which convolves with the layer kernel of size 3×3, single stride and having 64 filter channels at the first stage. Each convolutional layer extracts the image features and provides the map with depth size of $64 \times 2^{i-1}$, where $i = (1, 2, 3)$ is a layer index. The next max-pooling layer having kernel size is set to 2×2 with a stride of 2. After the third stage, concatenation of feature maps is performed and a 2-D feature vector is generated. This feature vector feds to the two-channel soft-max layer to compute the probability distribution in order to generate a final score map. For an input image of size $N \times M$, the size of the output feature map is $(N/2 - 2^i + 1) \times (M/2 - 2^i + 1)$, where $i = 3$ and the learning rate is initially set to 10^{-4}. In soft max layer, the logistic loss function of the output layer is minimized using a gradient descent approach to obtain the optimized weights and bias for each layer. The network contains two subsequent convolutional layers followed by max-pooling layer and the third convolutional layer as shown in Fig. 2.

Let r_k and $\widehat{r}_l$ are representing the kth input map and l^{th} output map of convolutional layer i, respectively. The non-linear formulation of the input and convolutional operator (weight matrix) can be expressed as,

$$\hat{r_l} = ReLU_f \left(\frac{bias_l + \sum_k convw_{k,l} * r_k}{\phi} \right) \tag{7}$$

where the $ReLU$ nonlinear activation function [24] can be expressed as $ReLU_f(\phi) = max(\phi, 0)$. The operator $*$ indicates the convolution operation, $convw_{k,l}$ indicates the convolutional kernel imposed on k^{th} input map to gain l^{th} output map, and $bias_l$ is the bias for each ith layer. The convolutional operation performed within the local mask of size $n \times n$ which is kept same for all three layers while the pooling layer reduces the resolution of the layer output by two.

In the proposed framework, CNN based base layer fusion is obtained in three key steps; 1) initial feature map generation using pre-trained CNN model, 2) activity measure and fusion using weighted average based rule, and 3) consistency verification of the fused image for adequate contrast and brightness in the resultant image. The detailed steps to fuse the base layer are explained in the next section.

C. Clustering PCA based Statistical Feature Mapping

This section focuses on the preservation and enhancement of structural features of an image based on patchwise statistical PCA clustering. This approach is composed of three basic steps as 1) patch partitioning based on geometrical details of the layers, 2) clustering the maximum variance regions and 3) learning the PCA basis for each class and combine each sub-basis to form a composite dictionary. For a given input data, the structural patch decomposition is applied based on statistical measures such as fine details, local content and the color variations. The decomposed patch groups are clustered using fuzzy c-means (FCM) to generate a different cluster groups for each measure. The fusion framework based on statistical clustering is shown in Fig. 3.

For an input image, an overlapped patch-based image partitioning is made with a fixed window and step size. After image partitioning, each patch is decomposed into two major components, namely local mean and detail structure of an image. The local mean μ is calculated by averaging pixels presented in patch p and representing the flat areas of p. The detail structure denoted by p_d is calculated by subtracting the patch mean from the extracted image patch p as follows,

$$p_d = p_{N \times M} - \mu_{N \times M} \tag{8}$$

With the use of the above defined components, each patch p is reconstructed as follows,

$$p = \bar{p}_d + \bar{\mu} \tag{9}$$

If the input image has three channels, then both the decomposed components are to be calculated for each color channel R, G, and B, separately and then each patch is reconstructed as,

$$p = \begin{bmatrix} \bar{p}_d^r \\ \bar{p}_d^g \\ \bar{p}_d^b \end{bmatrix} + \frac{1}{3}(\bar{\mu}_d^r + \bar{\mu}_d^g + \bar{\mu}_d^b) \tag{10}$$

where the first component represents the color structure extracted from the three channels and the second component

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represents the averaged local mean of the three channel color patches.

In the next phase, patch clustering is used to capture the pixels in one cluster based on structural similarity. For clustering the pixels, FCM based learning approach is utilized. Each cluster is defined through subspace basis constructed using PCA to get the prominent structures of the input data. Finally, the generated PCA basis representing the most common and significant features in detail image is applied to transform the image patches into PCA subspace [25].

III. Proposed approach

This section summarizes the implementation steps of the proposed method. A complete flow process is shown in Fig. 4. For the given input image pairs $\mathbf{R} = R(x, y)$ and $\mathbf{S} = S(x, y)$, the proposed fusion model is implemented as described below.

Implementation steps

Step 1: Covert the RGB model of an input image into YIQ color space using the given expression. [26].

$$\begin{bmatrix} Y \\ I \\ Q \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ 0.5959 & -0.2746 & -0.3213 \\ 0.2115 & -0.5227 & 0.3112 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} \tag{11}$$

where Y refers to luminance, I and Q refer to chromaticity based in-phase (cyan-red axis) and quadrature-phase (magenta-green) components, respectively.

Step 2: Decompose the luminance component (Y_X) of reference image $X \in (R, S)$ using hybrid two-scale layer decomposition into one base layer B_2^X and two detail layers D_1^X and D_2^X.

Step 3: **Base layer fusion-** A convolutional neural network based fusion approach is used to merge the base layer components from each decomposed data. To implement such an approach, a pre-trained CNN model given in [18] is used and a feature map is obtained at output layer. For each input, feature map is optimized using feature-based activity score map followed by consistency verification. The detailed fusion process is given in steps below:

1) Consider the base layer contents B_2^R and B_2^S which use a three layer CNN model to extract the activity measure for content mapping. For the ith convolutional layer and c channels of each layer, the output map can be expressed as

$$O_X^{i,c} = ReLU_{f^i}(\phi(B_2^X)), X \in (R, S) \tag{12}$$

where at each ith layer $\phi(\cdot)$ denotes the feature map acquired from Eq. (7). The first layer uses B_2^X as input image patch while the later stages use the $(i-1)^{th}$ layer output as the input map for the ith layer feature extraction denoted by $O_X^{i,c}$.

2) Calculate the initial weight map $W_X(y, z)$ at 2-D softmax layer using a concatenated output feature maps O_X from the third convolutional layer and expressed as,

$$W_X(y, z) = \frac{O_X(y, z)}{O_R(y, z) + O_S(y, z)} \tag{13}$$

3) Calculate the fused weight map using block-based regional energy (RE) as an activity measure for individual output maps and a weighted average is taken with upsampling to get a final activity similar in size with the reference input.

$$map_X(y, z) = RE_b(W_X(y, z)) \tag{14}$$

where RE refers to a weighted regional energy and the region is referred to a size of the weighted window $w_b(y, x)$ for the neighbourhood processing as given below.

$$RE_b(y, z) = \sum_{(j,i) \in w_b} w_b(j, i) W(j, i)^2 \tag{15}$$

where width $b = 2l + 1$, $y - l \leq j \leq y + l$, $z - l \leq i \leq z + l$ and the normalized weight value is defined in 5×5 neighbouring pixels for efficient feature masking.

$$w_b = \frac{1}{256} \begin{bmatrix} 1 & 4 & 6 & 4 & 1 \\ 4 & 16 & 24 & 16 & 4 \\ 6 & 24 & 36 & 24 & 6 \\ 4 & 16 & 24 & 16 & 4 \\ 1 & 4 & 6 & 4 & 1 \end{bmatrix} \tag{16}$$

$$\overline{map}_X(y, z) = \frac{1}{9} \left[\sum_{t=-1}^{1} \sum_{u=-1}^{1} map_X(y + t, z + u) \right] \tag{17}$$

$$W_{A_X}(y, z) = \frac{\overline{map}_X(y, z)}{\overline{map}_R(y, z) + \overline{map}_S(y, z)} \tag{18}$$

$$\tilde{W_X}(y + a, z + a) = W_{A_X}(y, z) \tag{19}$$

where b and a refers to block size and upsample unit, respectively. Block size $b = 5$ is considered here to preserve the useful high frequency features (i.e. edges) mapped from the CNN with required artifacts suppression. The upsampling step size is set as $a = 8$ to reconstruct the initial feature mapped and resized to the source image using the relation $a = (0, 1, ., ., ..2^{i-1} - 1)$, where i is layers in convolutional network. Finally, combine the resultant activity maps to preserve most of the prominent contents (brightness and structural information) relative to the two base layer coefficients as follows,

$$W_F(y, z) = \tilde{W_R}(y, z) \times B_2^R + \tilde{W_S}(y, z) \times B_2^S \tag{20}$$

4) *Post-processing of decision map* - To get the optimized decision map, post-processing is performed in three steps, namely (1) initial segmentation map using binary thresholding with threshold $median(W_F(y, z))$ to acquire binary segmented map (2) morphological filtering to reduce misclassified pixels (smoothen tiny areas) using bwareopen and close operation and (3) weighted guided filtering based weight map optimization. The final decision map improves spatial consistency and suppresses noise and artifacts. Let for the target segmented image $t(y, z)$ getting after step 2, the fused guidance

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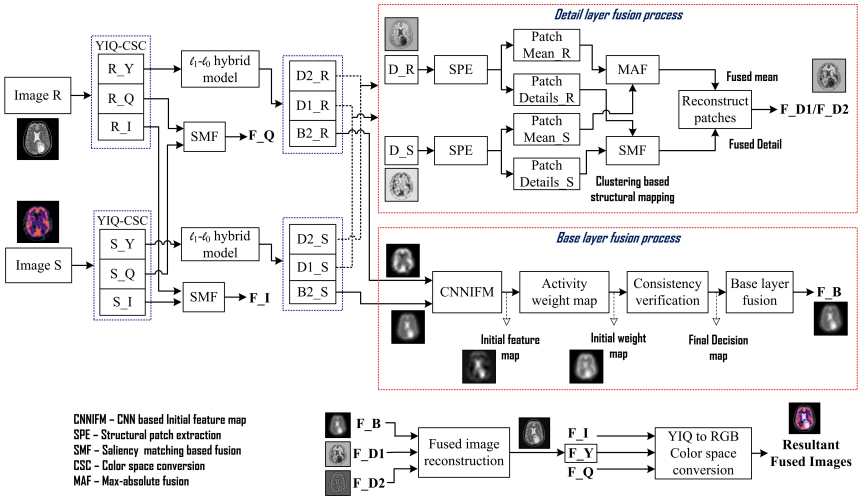


Fig. 4. Process flow of the proposed method

image $g(y,z)$, optimized final decision map $D(y,z)$, and fused base layer image B_2^F is obtained as given below.

$$g(y,z) = B_2^R \times t(y,z) + B_2^S \times (1 - t(y,z)) \quad (21)$$

$$D(y,z) = G_{r,\epsilon}(g(y,z), t(y,z)) \quad (22)$$

$$B_2^F = D(y,z) \times B_2^R + (1 - D(y,z)) \times B_2^S \quad (23)$$

where $G_{r,\epsilon}$ is the weighted guidance filter (WGF) [21] with filter radius (r) and degree of smoothing (ϵ), respectively.

Step 4: **Detail layer fusion -** For the detail layer fusion, the concept of statistical clustering based PCA [25] is employed, which improves the contrast level significantly and enhances the overall image clarity. The following procedures are considered to implement the fusion process discussed as,

1) *PCA basis learning* - Generate a compact PCA basis using an offline learning process to boost the computational efficiency. The training stage is done with 130 medical images of size 256×256. A sliding-window based patch extraction (with patch size 8 and stride 1) is applied to the training dataset and two structural components (patch mean and patch details) are computed for each patch. An FCM based structural clustering is used to index the patch in N groups having almost similar features but different intensities. For each cluster, cluster centroid and PCA based approximation model are computed by choosing the uppermost principal components. The resultant principal subspaces are combined to generate a dictionary P, consisting of major elements of the clustered patch.
2) Partition the detail layer components D_i^X into overlapped patches (size 8 and step 1) p_i^X, where $X \in (R,S)$ and $i \in (2,1)$ is the decomposition scale.
3) For each patch, compute the statistical information as patch mean and patch structure similarly expressed in Section II and denoted by μ_i^X and p_i^X , respectively.
4) Compute the class of each patch based on minimum Euclidean distance between the centroids of each cluster and patch $p_i^{dX}(y,z)$. Apply the learned PCA dictionary on patch component p_i^{dX} to extract the most relevant local structures s_i^{dX} as given below.
$$s_i^{dX} = P \cdot p_i^{dX} \quad (24)$$
The larger value of s_i^{dX} defines the more detail information, while the smaller define the fine texture details.
5) Reconstruct the approximated patch detail model $\tilde{p}_i^{dX}$ using,
$$\tilde{p}_i^{dX} = P^T \cdot s_i^{dX} \quad (25)$$
6) Fuse the patch mean using choose-max(abs) rule as,
$$\mu_i^F = \begin{cases} \mu_i^R & |\mu_i^R| > |\mu_i^S| \\ \mu_i^S & otherwise \end{cases} \quad (26)$$
7) Fuse the patch structural information using the saliency measure based on local contrast variation to preserve the edge details followed by matching the saliency measures. For two inputs $\tilde{p}_i^{d_R}$ and $\tilde{p}_i^{d_S}$, the required computation are expressed as,
$$\tilde{p}_i^{d_F} = d_i(y,z) \cdot \tilde{p}_i^{d_R} + (1 - d_i(y,z)) \cdot \tilde{p}_i^{d_S} \quad (27)$$
where $d_i(y,z)$ and $LC_i^X(y,z)$ represent a decision weight map and local contrast based patch saliency measure, respectively. $M_i(y,z)$ represents a match measure.

$$d_i(y,z) = \begin{cases} \tau, & M_i(y,z) \le \tau, LC_i^R(y,z) > LC_i^S(y,z) \\ 1-\tau, & M_i(y,z) \le \tau, LC_i^R(y,z) \le LC_i^S(y,z) \\ 0.5 + 0.5\left(\frac{1-M_i(y,z)}{1-\tau}\right), & M_i(y,z) > \tau, LC_i^R(y,z) > LC_i^S(y,z) \\ 0.5 - 0.5\left(\frac{1-M_i(y,z)}{1-\tau}\right), & M_i(y,z) > \tau, LC_i^R(y,z) \le LC_i^S(y,z) \end{cases} \quad (28)$$

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$$LC_i^X(y,z) = \sum_{w\in 3} \left| \tilde{p}_i^{d_X}(y+w, z+w) \right|^2 \tag{29}$$

and

$$M_i(y,z) = \frac{\begin{bmatrix} 2\sum \tilde{p}_i^{d_R}(y+w, z+w) * \\ \tilde{p}_i^{d_S}(y+w, z+w) \end{bmatrix}}{\sum_{X\in(R,S)} LC_i^X(y,z)} \tag{30}$$

where $\tilde{p}_i^{d_F}$ denotes the fused structural patch detail layer $i \in (1,2)$. τ is a visual threshold value to control the saliency-based match measure mapped between two different source images. The aforementioned formulations present an inter-dependency between the $LC(y,z)$ and $M(y,z)$ followed by a decision weight map $d(y,z) \in [0,1]$. If the local contrast based pattern similarity is higher than the visual threshold (τ), then the decision weight map will vary nearly about ± 0.5 based on the dominant patterns between two images. However, if the similarity is lower than the visual threshold at a given spatial position or the salient patterns are less correlated, the image having more significant saliency will be weighted more typically by $d_{max}(y,z) = \tau$ in the decision map. In the proposed work, the parameter $\tau = 0.7$ is considered to generate the best possible decision weight map according to Burt's method [27].

8) Combine the fused patches from step 6 and 7 to reconstruct the fused detail layers as given

$$D_i^F = \tilde{p}_i^{d_F} + \mu_i^F \tag{31}$$

Step 5: Reconstruct the resultant fused luminance image using three fused layers and can be expressed as

$$F^Y = B_2^F + D_2^F + D_1^F \tag{32}$$

Step 6: Reconvert the fused image to RGB color space using YIQ to RGB transformation using the expression given below and get the fused outcome.

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1 & 0.956 & 0.621 \\ 1 & -0.272 & -0.647 \\ 1 & -1.106 & 1.703 \end{bmatrix} \begin{bmatrix} Y \\ I \\ Q \end{bmatrix} \tag{33}$$

The salient steps of the proposed fusion methods are summarized in Algorithm 1.

IV. Experimental Details

To evaluate the performance of the proposed method, seventy three image pairs are used (14 CT-MR, 29 MR-SPECT, and 30 MR-PET images). The testing images taken from Harvard whole brain atlas are pre-registered. To implement the proposed approach, a hybrid norm based two-scale layer decomposition is employed with CNN for base layer fusion and statistical clustering for detail layer fusion.

Two-scale layer decomposition technique requires the appropriate settings of three parameters namely λ_1, λ_2, and λ_3 within the range of $[0,1]$. The degree of smoothness of the first layer is controlled by λ_1 and λ_2 for the base and detail layer obtained at scale 1, respectively. However, the second scale has an only λ_3 that controls the smoothing level of the base layer obtained at scale 2. The regularization parameter λ_1, and λ_2 are set as 0.4, $0.01\lambda_1$ respectively. λ_3 is set to 0.05 as it controls the second scale base layer and due to low accuracy of medical images, it kept to a smaller value than the layer one. In the base layer fusion, CNN based initial decision map is optimized using the WGF controlled by parameters radius r and blurring degree ϵ and their values are set as 11 and 0.01, respectively.

To assess the performance analysis of the fusion results, the proposed method is compared with several existing fusion methods such as WT based fusion proposed by Nirmla et al., 2016 [28], PCNN based fusion on NSCT domain proposed by Das and Kundu, 2012 [9], guided filtering (GFF) based fusion by Li et al., 2013 [16], NSST domain PCNN based fusion proposed by Ganasala and Kumar, 2016 [29], fuzzy adaptive PCNN based fusion in NSCT domain by Yang et al., 2017 [11], gradient domain guided filtering (GDGF) based fusion by Zhu et al., 2018 [30], CNN based medical image fusion by Liu et al., 2017 [18], cascaded fusion framework by Singh et al., 2018 [12], CNN based fusion by Hermessi et al. [17] and NSST domain CNN model by Hou et al. [31]. Moreover, other approaches developed by Bhatnagar et al., 2013 [8], Yang et al., 2015 [32], Yang et al. (NSCT-Type2 Fuzzy), 2016 [33] are considered for color image fusion.

A. Objective Assessments

To investigate the efficacy of the proposed method, different performance measures are selected to present the comparative evaluation. The subjective results speculate the comparative assessment of visual perception among the source and fused images while the quantitative investigation demonstrates the usefulness of the perceived results through various quality measures. These measures are considered as entropy (En), mutual information (MI), standard deviation (STD), universal image quality index (UIQI), feature mutual index (FMI), spatial frequency (SF) and the Xydeas edge index (XEI) stated in Table I. Higher index values show a better fusion performance for all considered objective measures.

B. Experimental Results and Discussion

1) Comparison of Anatomical-Anatomical Image Fusion Results

This section discusses the different simulation methods and their results for the presented framework. In the proposed work, four sets of the multimodal dataset are taken to evaluate the fusion performance. The set-1 shown in Fig. 5 includes four different pairs of CT in Fig. 5(a) and MR-T2 in Fig. 5(b) images with their fused results obtained by the proposed method shown in Fig. 5(c). The quantitative measures also demonstrate the higher values of En, STD, MI, SF, and XEI for presenting a better fusion accuracy.

Furthermore, to validate the effectiveness of the proposed scheme, a comparative analysis is made with ten existing fusion methods. The fusion results of different CT and MR image pairs are shown in Fig. 6. Furthermore, a zoomed view has also shown to judge the subjective performance of each method more clearly. In Fig. 6(g), CNN based fusion

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Algorithm 1: Proposed Method

Input: Pre-registered image pair $R^{M\times N}$ and $S^{M\times N}$, transform dictionary P, centroid (learned from training samples) and CNN model
Output: Fused image
Parameters: λ_1, λ_2, λ_3 (decomposition phase), r and ϵ (decision map optimization), patch size and stride (sliding-window based detail image partitioning)

1) Convert the images R and S from RGB space to YIQ color space and take Y_X, $X = (R, S)$ component for further steps.
2) Decompose Y_X component into B_2^X, D_2^X and D_1^X using Eq. (1)-(5).
3) **Base layer fusion**
   - Load the pre-trained CNN model for initial feature mapping O_X.
   - Compute the two-class weight map W_X using soft-max layer.
   - Compute energy based the regional activity map of W_X and upsample it to obtain final activity map $\tilde{W}_X$.
   - Compute initial decision map using Eq. (20).
   - Optimize the decision map $D(y, z)$ using consistency verification step.
   - Compute fused base layer using Eq. (21)-(23).
4) **Detail layer fusion**
   - Partition D_i^X into overlapped patches p_i^X for each scale $i = (1, 2)$.
   - For each patch, p_i^X computes patch mean μ_i^X and patch details p_i^{dX}.
   - Compute minimum Euclidean distance between centroid and p_i^{dX} to get cluster assignment for each p_i^{dX}.
   - **Loop** $j = 1, 2, ..., max(assignment)$
     - Extract highlighted structure detail using Eq. (24).
     - Reconstruct approximated detail layer using Eq. (25).

     end
   - Fuse μ_i^X using $max(abs)$ rule and get μ_i^F.
   - Fuse $\tilde{p}_i^{dX}$ using saliency measure and consistency verification and get $\tilde{p}_i^{dF}$.
   - Aggregate patches to get reconstructed fused detail layer using Eq. (31).
5) Combine fused detail and base layers to get final fused output using Eq. (32).
6) Reconvert the YIQ color space into the RGB space.

TABLE I
EVALUATION PARAMETERS CONSIDERED FOR ANALYZING THE FUSION PERFORMANCE

Sl. No.	Evaluation parameters	Mathematical expressions
1.	Entropy (En)	$En = -\sum_{i=0}^{L-1} p\,(i)\, log_2 p\,(i)$ Higher entropy values refer to the more amount of information present in the fused image.
2.	Standard deviation (STD)	$STD = \sqrt{\sum_{i=1}^{M}\sum_{j=1}^{N} \frac{\left[X(i,j)-(\frac{1}{M\times N}\sum_{i=1}^{M}\sum_{j=1}^{N} X(i,j)\right]^2}{M\times N}}$ A higher value of STD indicates the higher contrast of the fused image compared to the source images.
3.	Mutual information (MI) [34]	$MI = I_{FR}(f; r) + I_{FS}(f; s)$ where $I_{FX}(f; x) = \sum_{f,x} p_{FX}(f, x) log_2 \frac{p_{FX}(f,a)}{p_X(x)p_F(f)}$ Higher MI values indicate the more amount of feature information in the fused image transferred from the source image.
4.	Universal image quality index (UIQI) [35]	$UIQI = \left(\frac{\sigma_{FX}}{\sigma_F\sigma_X}\right)\left(\frac{2\mu_F\mu_X}{\mu_F^2+\mu_X^2}\right)\left(\frac{2\sigma_F\sigma_X}{\sigma_F^2+\sigma_X^2}\right)$, where μ_X , μ_F and σ_X^2, σ_F^2 refer to the average and variance of the source and fused images, respectively.
5.	Spatial frequency (SF) [12]	$SF = \sqrt{RF^2 + CF^2}$, where $RF = \sqrt{\frac{1}{M(N-1)}\sum_{i=1}^{M}\sum_{j=2}^{N}(X(i, j-1) - X(i, j))^2}$ $SF = \sqrt{\frac{1}{N(M-1)}\sum_{i=2}^{M}\sum_{j=1}^{N}(X(i, j) - X(i-1, j))^2}$
5.	Feature mutual information (FMI) [34]	$FMI = \frac{I_{FR}}{H_F+H_R} + \frac{I_{FS}}{H_F+H_S}$, where $I_{FX} = \sum_{FX} p_{FX} log_2 \frac{p_{FX}}{p_F p_X}$ and H_X refer to the histogram based entropies of the corresponding image. Higher values of FMI indicate the higher regional mutual information between corresponding window size 3×3 in the fused and source images.
6.	Xydeas edge index (XEI) [36]	$Q^{RS/F} = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N} Q^{RF}(i,j)w^R(i,j)+Q^{SF}(i,j)w^S(i,j)}{\sum_{i=1}^{M}\sum_{j=1}^{N} w^R(i,j)+w^S(i,j)}$ $Q^{RF}(i, j) = Q_e^{RF}(i, j) + Q_o^{RF}(i, j)$, where $Q_e^{RF}(i, j)$ and $Q_o^{RF}(i, j)$ refer to the edge strength and orientation preservation values, respectively. Higher values indicate the higher degree of edge preservation.

provides a satisfactory fusion results but edge outlines are not seen clear in the resultant image. From Fig. 6(h), it can be clearly seen that the method is able to preserve the significant information with natural contrast, however, shows smoothed edge details in the area that has small but frequent variations. CNN based similarity learning approach [17] retains most of the feature information but has blurred contours near fine edges with over brightness as shown in Fig. 6(i). Fig. 6(j) present the fuse images obtained by Hou et al. [31] which is able to capture edge details and retain the overall information very well, however, limits to maintain the pixel contrast. The fusion results of the proposed method shown in Fig. 6(k) are characterizing principal and tiny details with required brightness and pixel contrast. In addition to this, it is also able to preserve sharp edges and visual consistency quite well and provides halo-free resultant image.

Furthermore, the fusion results of eight existing methods are also analyzed objectively to validate the corresponding subjective performances. Table II shows the objective fusion results for all four pairs. Their averaged scores are also shown in Table III. From Table II, it is clearly seen that the proposed method achieves highest edge index *XEI* and subsequent information from the source i.e. *MI*.

2) Comparison of Anatomical-Functional Image Fusion Results

In this section, five pairs of color medical images (MR-

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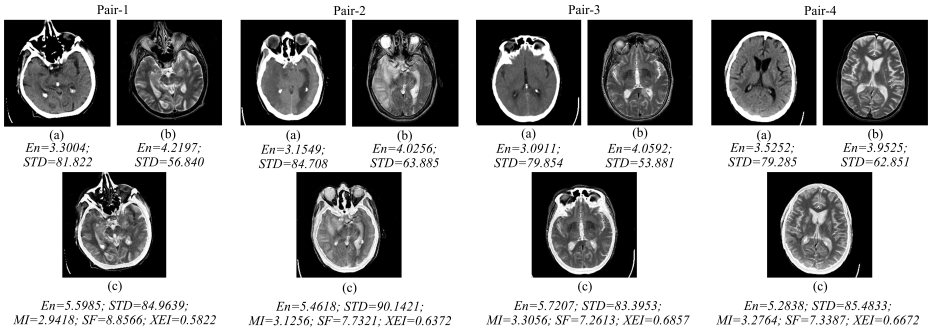


Fig. 5. Fusion performance evaluated by the proposed approach on (a) Source CT image (b) Source MR image (c) Fused images

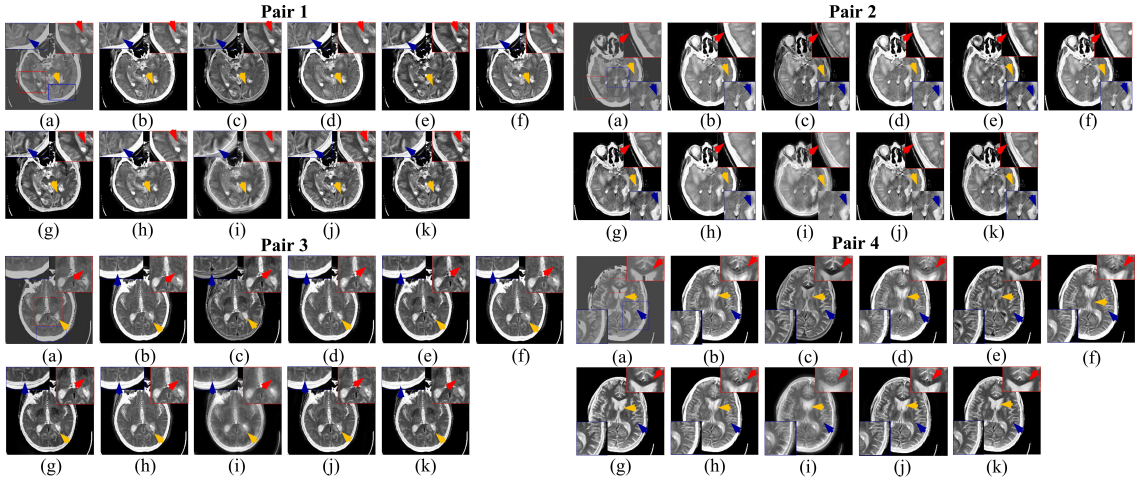


Fig. 6. Four sets of CT-MR image fusion results. Comparison of fused images obtained by (a) Nirmla et al., 2016 [28] (b) Das and Kundu, 2012 [9] (c) Li et al. (GFF), 2013 [16] (d) Ganasala and Kumar, 2016 [29] (e) Yang et al., 2017 [11] (f) Zhu et al. (HMSD-GDGF), 2018 [30] (g) Liu et al., 2017 [18] (h) Singh et al., 2018 [12] (i) Hermessi et al., 2018 [17] (j) Hou et al., 2019 [31] (k) Proposed method

TABLE II
COMPARATIVE ANALYSIS OF THE PERFORMANCE MEASURE EVALUATED FOR CT-MR IMAGE PAIRS

Fusion Methods	Pair 1					Pair 2				
	En	*STD*	*MI*	*SF*	*XEI*	*En*	*STD*	*MI*	*SF*	*XEI*
Nirmla et al. , 2016 [28]	5.076	60.803	2.6754	7.8695	0.4369	4.901	60.823	2.4851	7.1971	0.4043
Das and Kundu, 2012 [9]	5.3184	83.368	2.8223	7.3292	0.4866	4.9108	89.489	2.9532	6.6292	0.4729
Li et al. (GFF),2013 [16]	5.1317	63.433	1.0962	7.8214	0.4286	4.6649	66.054	1.3061	6.2101	0.4190
Ganasala and Kumar, 2016 [29]	5.2205	82.372	2.5842	5.8048	0.5114	4.8321	85.214	2.8754	6.2896	0.5308
Yang et al., 2017 [11]	5.2361	83.448	2.5875	7.5436	0.5375	4.8843	87.183	2.8417	6.5436	0.5547
Zhu et al. (HMSD-GDGF), 2018 [30]	5.2728	82.218	2.8077	7.7862	0.5013	5.0989	88.006	2.8441	7.0792	0.5681
Liu et al (CNN), 2017 [18]	5.1036	79.469	2.6576	8.3656	0.5675	4.7416	84.192	2.7226	7.7298	0.6298
Singh et al., 2018 [12]	5.5147	84.887	2.8824	8.8542	0.5346	5.2241	**91.124**	3.0874	**7.7845**	0.5584
Hermessi et al., 2018 [17]	5.1254	71.79	2.5340	7.5836	0.4336	5.2938	81.122	2.6920	6.9385	0.5021
Hou et al., 2019 [31]	5.5137	80.064	2.6633	8.3548	0.4958	5.3007	84.757	2.7757	7.7423	0.5439
Proposed	**5.5985**	**84.964**	**2.9418**	**8.8566**	**0.5822**	**5.4618**	90.142	**3.1256**	7.7321	**0.6372**
Fusion Methods	Pair 3					Pair 4				
	En	*STD*	*MI*	*SF*	*XEI*	*En*	*STD*	*MI*	*SF*	*XEI*
Nirmla et al. , 2016 [28]	4.7054	60.823	2.7542	6.7467	0.4104	4.6585	60.926	2.7941	6.3218	0.4318
Das and Kundu, 2012 [9]	4.9361	81.587	3.0489	6.6242	0.5088	4.724	84.361	3.0406	6.7005	0.4203
Li et al. (GFF), 2013 [16]	4.6573	58.294	1.4823	6.4681	0.474	4.5797	62.779	1.4205	6.8758	0.4616
Ganasala and Kumar, 2016 [29]	4.7692	79.859	3.0513	5.8048	0.5328	4.7512	80.108	3.0113	5.6939	0.5050
Yang et al., 2017 [11]	4.8087	81.464	2.9881	6.5477	0.5461	4.7173	81.997	2.9531	5.8631	0.5046
Zhu et al. (HMSD-GDGF), 2018 [30]	4.8141	80.935	3.0815	6.8501	0.5418	4.5953	83.695	3.0517	6.8237	0.5388
Liu et al (CNN), 2017 [18]	4.8860	76.170	2.8743	7.2448	0.6533	4.871	81.716	3.0315	7.2616	0.6635
Singh et al., 2018 [12]	5.1623	82.784	3.2654	6.9098	0.5682	5.1422	85.263	3.1847	6.9684	0.5086
Hermessi et al., 2018 [17]	5.2991	72.629	2.8224	6.6074	0.5108	5.3659	75.363	2.8634	6.490	0.4524
Hou et al., 2019 [31]	5.208	78.576	2.9878	7.4378	0.5729	5.3873	79.650	2.9672	7.6164	0.5511
Proposed	**5.7207**	**83.395**	**3.3056**	**7.2613**	**0.6857**	**5.2838**	**85.488**	**3.2764**	**7.3387**	**0.6672**

SPECT/PET) are considered to assess the fusion results of the proposed method and a comparative analysis is made. From Fig. 7(a) and (b), pair-1, 2, and 3 depict MR-T2 and 99mTc-SPECT images, while pair-4 and 5 show MR-T2 and PET-FDG images, respectively. The fused images obtained by the proposed method are presented in Fig. 7(c) which shows a better quality image with the significant structural and color content captured from the input images. To validate the performance of the proposed method, the quantitative measures are compared with existing state-of-the-art methods shown in Table IV. The comparison of fusion results obtained by the proposed and other methods is shown in Fig. 8. Fig. 8(c) demonstrates the work proposed by Bhatnagar et al. [8] which shows a good amount of luminance restoration, but the edge detail gets completely blurred. The objective parameters also indicate a lower value of *FMI* and *SF*. In Fig. 8(d), GFF

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TABLE III
AVERAGED EVALUATION MEASURES (MEAN ± STANDARD DEVIATION) FOR CT-MR NEUROLOGICAL IMAGES

Fusion Methods	Performance Measures				
	En	*STD*	*MI*	*SF*	*XEI*
Source MR	3.8803 ± 0.3366	59.3509 ± 4.9843	-	6.3309 ± 0.8517	-
Source CT	3.0647 ± 0.3035	83.9079 ± 5.4953	-	5.2680 ± 0.6784	-
Nirmla et al. , 2016 [28]	4.8122 ± 0.1812	60.8751 ± 0.0445	2.9276 ± 0.2097	7.0909 ± 0.4656	0.4505 ± 0.1289
Das and kundu, 2012 [9]	4.8081 ± 0.3713	86.9047 ± 4.4341	3.0086 ± 0.1102	6.2976 ± 0.6017	0.4785 ± 0.0490
Li et al. (GFF), 2013[16]	4.6501 ± 0.2949	64.0659 ± 3.7542	1.3093 ± 0.2232	6.3345 ± 0.6796	0.4458 ± 0.0530
Ganasala and Kumar, 2016 [29]	4.9135 ± 0.2426	84.0917 ± 4.3443	2.8656 ± 0.1751	5.8675 ± 0.4616	0.5199 ± 0.0513
Yang et al., 2017 [11]	5.0273 ± 0.2126	84.9058 ± 4.3424	2.9026 ± 0.2145	6.4654 ± 0.6297	0.5477 ± 0.0436
Zhu et al. (HMSD-GDGF), 2018 [30]	4.9127 ± 0.3124	83.1357 ± 3.1570	2.9225 ± 0.1497	7.1182 ± 0.4261	0.5384 ± 0.0448
Liu et al (CNN), 2017 [18]	4.7219 ± 0.2021	80.6907 ± 3.2741	2.7915 ± 0.1742	7.2672 ± 0.6489	0.6118 ± 0.1451
Singh et al., (2018) [12]	5.1529 ± 0.2378	88.5082 ± 4.4684	3.1594 ± 0.1637	7.1657 ± 1.1289	0.5578 ± 0.0465
Hermessi et al., 2018 [17]	5.3857 ± 0.3374	75.7403 ± 3.3521	2.7978 ± 0.1726	7.1092 ± 0.4521	0.4954 ± 0.0403
Hou et al., 2019 [31]	5.467 ± 0.3760	81.2753 ± 2.055	2.9184 ± 0.1784	7.4922 ± 0.3891	0.5616 ± 0.0324
Proposed	5.6749 ± 0.2554	89.9548 ± 4.2868	3.4304 ± 0.1244	7.5627 ± 0.6894	0.6559 ± 0.0419

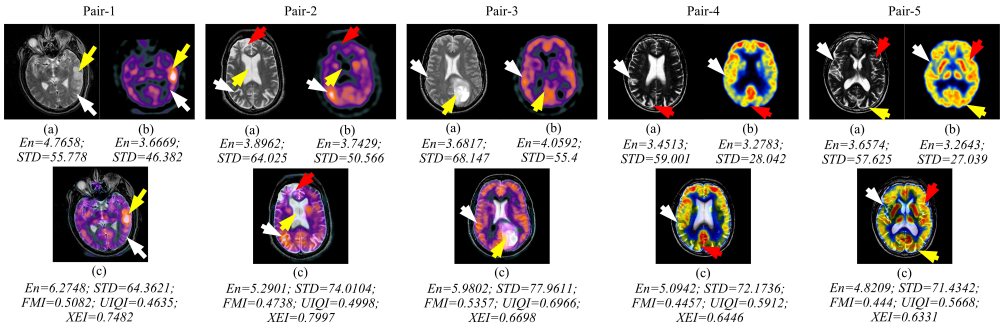


Fig. 7. Fused images using proposed fusion method on various test neurological images along with quantitative values metrics. Pair 1, 2 and 3 input images (a) MR image (b) SPECT image (c) Fused image, Pair-4 and 5 Input image (a) MR image (b) PET image (c) Fused image

method preserves the spatial information well but limited to poor spectral variation and proportions. In Fig. 8(e), fusion in NSCT domain nicely captured the abnormal tissue details with good pixel contrast, but the structural details shown by the red arrow have become visually less perceptible. Another fusion approach [33] gains significant information, but the tissue area shown by the yellow arrows suffers from poor spatial contrast as shown in Fig. 8(f). In Fig. 8(g), the fusion approach shows better spatial consistency, but lacks few basic anatomical details such as some focal lesions having different information and morphology. Fig. 8(h) achieves a good amount of average informative content and spatial consistency. However, some tiny edges appeared less sharp in the salient regions. The visual results are also validated by the poor value of *STD* and *XEI* metrics. Fig. 8(i) shows clearly the preservation of structural content while the fine edges are blurred and inconsistent in the fused image. Fig. 8(j) shows the overall detail information in the fused images with moderated *SF* values but limited by poor contrast and lower *STD*. The fused images obtained by the proposed method are shown in Fig. 8(k) that shows the ability to extract the object and edge detail without incorporating any artifact. The fine lines near the boundary area are clearly visible with appropriate sharpness. Furthermore, the objective analysis shows that the proposed framework is able to capture the significant visual features with higher *FMI* score and can preserve the significant luminance and structural information (higher *En* and *XEI* values).

The averaged comparative analysis is tabulated in Table V and shows the overall fusion performance which indicates that the fusion outcomes obtained by the proposed method contain most of the information with appropriate spatial consistency and spectral balance. The fused results also retain the high degree of structural similarity and able to mutually correspond to the source images.

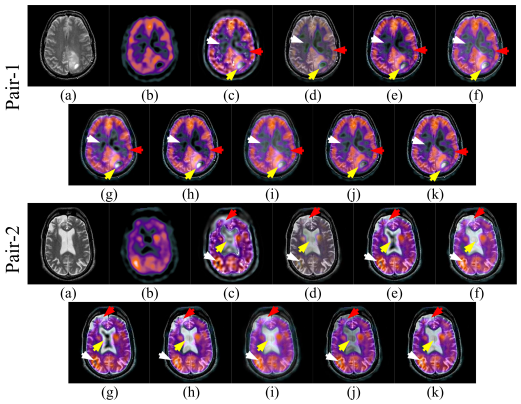


Fig. 8. Input image (a) MR image (b) SPECT image, qualitative comparisons of fused images obtained by (c) Bhatnagar et al., 2013 [8] (d) Li et al. 2013 [16] (e) Yang et al., 2015 [32] (f) Yang et al. 2016 [33] (g) Yang et al., 2017 [11] (h) Liu et al., 2017 [18] (i) Hermessi et al., 2018 [17] (j) Hou et al., 2019 [31] (k) Proposed Method

3) Computational Complexity Analysis

In this section, the comprehensive algorithm complexity and computational time of the proposed method are compared with the eleven different fusion methods and given in Table VI. All the experiments have been performed using MATLAB 2014a and a machine having Intel^{TM} Core2Duo 2.93 GHz processor with 8 GB RAM on Windows 10. The computational complexity for the proposed framework is explained as follows.

The number of computations required for the layer decom-

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TABLE IV
COMPARATIVE ANALYSIS OF THE PERFORMANCE MEASURE EVALUATED FOR MR-SPECT IMAGE PAIRS TAKEN BY DIFFERENT FUSION FRAMEWORK

Fusion Methods	Pair 1						Pair 2					
	En	*STD*	*FMI*	*SF*	*UIQI*	*XEI*	*En*	*STD*	*FMI*	*SF*	*UIQI*	*XEI*
Bhatnagar et al., 2013 [8]	4.5028	64.4275	0.2072	6.2531	0.3186	0.1709	4.6860	65.4292	0.2008	6.2755	0.2558	0.1443
Li et al. (GFF), 2013 [16]	4.5323	53.3025	0.3418	6.4931	0.4083	0.6036	4.5575	59.5072	0.3805	6.8999	0.3455	0.6031
Yang et al., 2015 [32]	4.3769	65.4121	0.4212	7.2324	0.3780	0.6455	4.5983	71.9412	0.4440	7.3450	0.3321	0.7153
Yang et al. , 2016 [33]	5.0614	64.7547	0.3157	6.7829	0.4282	0.4318	5.2762	72.6751	0.2778	6.9753	0.3891	0.4912
Yang et al., 2017 [11]	4.5904	64.0905	0.4005	7.3659	0.5163	0.6314	4.7683	68.9441	0.4365	7.5129	0.3986	0.6960
Liu et al. (CNN), 2017 [18]	4.8457	62.0423	0.4230	6.9828	0.5314	0.6347	4.8582	69.6076	**0.4762**	7.0191	0.4047	0.7036
Hermessi et al., 2018 [17]	**5.1564**	63.0452	0.2567	5.9659	0.3869	0.3936	5.2195	70.5628	0.2396	5.7947	0.3560	0.4633
Hou et al., 2019 [31]	5.0719	58.4451	0.2543	6.6938	0.3836	0.4047	5.0865	59.9375	0.2362	6.795	0.3607	0.3725
Proposed Method	5.1298	**65.8217**	**0.4986**	**7.6333**	**0.6337**	**0.6564**	**5.2901**	**74.0104**	0.4738	**7.6118**	**0.4998**	**0.7997**

TABLE V
AVERAGED EVALUATION MEASURES (MEAN $\pm$ STANDARD DEVIATION) FOR MR-SPECT NEUROLOGICAL IMAGES

Fusion Methods	Performance Measures					
	En	*STD*	*FMI*	*SF*	*UIQI*	*XEI*
Bhatnagar et al., 2013 [8]	4.3566 $\pm$ 0.6876	65.146 $\pm$ 7.5963	0.1948 $\pm$ 0.0135	5.7691 $\pm$ 0.7218	0.2455 $\pm$ 0.0495	0.1394 $\pm$ 0.0253
Li et al. (GFF), 2013 [16]	4.3049 $\pm$ 0.6636	57.343 $\pm$ 9.1360	0.3855 $\pm$ 0.0515	7.0841 $\pm$ 0.7412	0.3777 $\pm$ 0.0607	0.6648 $\pm$ 0.0621
Yang et al., 2015 [32]	4.3564 $\pm$ 0.6686	68.209 $\pm$ 9.0919	0.4493 $\pm$ 0.0348	7.0941 $\pm$ 0.8285	0.3376 $\pm$ 0.0598	0.6879 $\pm$ 0.0621
Yang et al. , 2016 [33]	4.8688 $\pm$ 0.6712	69.114 $\pm$ 7.2812	0.2593 $\pm$ 0.0379	6.9678 $\pm$ 0.5698	0.4165 $\pm$ 0.1227	0.6169 $\pm$ 0.1221
Yang et al., 2017 [11]	4.5966 $\pm$ 0.6623	67.262 $\pm$ 8.5285	0.4447 $\pm$ 0.0349	7.2612 $\pm$ 0.6741	0.4020 $\pm$ 0.0404	0.6842 $\pm$ 0.0645
Liu et al. (CNN), 2017 [18]	4.6984 $\pm$ 0.6372	65.509 $\pm$ 6.4233	0.4619 $\pm$ 0.0377	7.2421 $\pm$ 0.6006	0.4454 $\pm$ 0.1363	0.4166 $\pm$ 0.1206
Hermessi et al., 2018 [17]	5.7734 $\pm$ 0.3598	68.3259 $\pm$ 5.8216	0.2374 $\pm$ 0.0326	6.1468 $\pm$ 0.2660	0.3175 $\pm$ 0.0952	0.3234 $\pm$ 0.1010
Hou et al., 2019 [31]	5.0227 $\pm$ 0.5112	56.0334 $\pm$ 7.5614	0.2253 $\pm$ 0.0259	6.8656 $\pm$ 0.5813	0.3282 $\pm$ 0.0896	0.3498 $\pm$ 0.0954
Proposed Method	5.2184 $\pm$ 0.5908	69.385 $\pm$ 7.7885	0.4953 $\pm$ 0.0377	7.5582 $\pm$ 0.5778	0.4928 $\pm$ 0.1175	0.7306 $\pm$ 0.1235

TABLE VI
COMPARISON OF AVERAGED RUNNING TIME (MEAN $\pm$ STANDARD DEVIATION) TAKEN BY DIFFERENT FUSION FRAMEWORKS

Fusion Methods	Running time (sec)
Nirmla et al., 2010 [28]	0.5852 $\pm$ 0.0265
Das and Kundu., 2012 [9]	62.909 $\pm$ 4.6848
Li et al. (GFF), 2013 [16]	0.4552 $\pm$ 0.0507
Yang et al., 2015 [32]	7.8319 $\pm$ 0.5935
Ganasala and Kumar, 2016 [29]	6.1528 $\pm$ 0.4892
Yang et al. (NSCT-Type2 Fuuzy), 2016 [33]	10.161 $\pm$ 0.5842
Yang et al., 2017 [11]	89.282 $\pm$ 6.7748
Zhu et al. (HMSD-GDGF), 2018 [30]	28.313 $\pm$ 0.6421
Liu et al. (CNN), 2017 [18]	20.511 $\pm$ 0.0341
Hermessi et al., 2018 [17]	466.57 $\pm$ 15.521
Hou et al., 2019 [31]	42.096 $\pm$ 6.914
Proposed method	19.312 $\pm$ 0.0219

position method is $O(Nlog(N))$. The base layer and detail layer fusion process are taking approximately 15 sec and 4 sec, respectively. The averaged execution time is also listed in Table VI. From Table VI, it is clearly visualized that the proposed method moderately performed in terms of time complexity and takes about 19$\sim$20 sec. The parameter settings for the implementation of other methods are listed in Table VI are considered as mentioned in their respective papers.

V. CONCLUSION

In this paper, a unified fusion framework based on two-layer norm decomposition and feature learning is presented. The method highlights the ability to preserve the layer information (structural, fine details, brightness, and color) and increases the fusion accuracy. In the proposed method, a novel ℓ_n-norm based two-scale hybrid layer decomposition method is used to separate out the main information from the textural details. Based on visual sensitivity of the HVS model, most prominent features are extracted from both the base and detail layer and then combined based on feature preserving fusion rules. The experimental results show that the proposed method can efficiently extract the complex structures and maintains the spectral information as well without introducing any processing artifacts. The comparative results demonstrate that the proposed method shows superiority to many recent works in terms of the preservation of significant information with improved contrast and overall fusion quality of multimodal images.

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